



GROUND-N1 AND GROUNDNODE

Simulation User Manual

How to run, explore and present the city simulation and the Ground-N1 demo: an air taxi and the ground talking to each other, with five landing pads.

City simulation · Ground-N1 launcher, pilot, ground control, live view · Guided encounter · Aircraft · Smart pad · Integration lab

1. What this is

The project simulates an urban air mobility terminal with **five landing pads (GN-01 to GN-05)**. It has two parts you can use on their own or together:

- **City simulation (the home page).** Air taxis (UAMs) and small drones (UAVs) fly out from the pads, hover still over a city site, return, land and charge. Use it to show the whole fleet at work.
- **Ground-N1 demo.** One air taxi and the ground node talk to each other. A pilot uses a cockpit screen, the ground team uses an operations screen, and a live 3D view shows the pads. Every instruction is signed and checked, and the ground computer explains each decision.

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2. Getting started

1. Install **Node.js** (any recent version) and open a terminal in the project folder.
2. Run `npm install` once. This downloads the pinned 3D library.
3. Run `npm start`. It serves the project at `http://localhost:8123`.
4. Open `http://localhost:8123` in **Chrome or Edge**. The home page is the city simulation.

Check it works: run `npm test`. Every suite should print a PASS line (city fleet, protocol, coordination, encounter, Ground-N1 and the site checks).

Requirements

- A desktop browser with WebGL (Chrome, Edge or Firefox). The three Ground-N1 windows must be in the **same browser** so they can share one simulation.
- For the two-screen demo, at least 1280 × 720 per window. A second monitor is ideal for the live view.
- By default the server is reachable only from this computer. To show it on your network run `HOST=0.0.0.0 npm start`.

3. Finding your way around

Every page has the same header. Click the **Vahnim logo** or **Home** to return to the city simulation from anywhere.

Header item	What is inside
Home	The city simulation.
Ground-N1 demo	Sub-menu: Launcher, Pilot, Ground control, Live view, Demo wall.
Explore	Sub-menu: Overview, Guided encounter, Aircraft, Smart pad, Integration lab.
Guide ↗	Opens this manual (PDF).
Help	A short plain-language explanation of the page you are on. The button pulses until you have read it once. Press <code>Esc</code> to close it.

Clicking a header item replaces the current window. During a Ground-N1 demo keep the Pilot, Ground and Live windows open; the shared simulation keeps running while at least one is open and restarts if you close them all.

4. City simulation

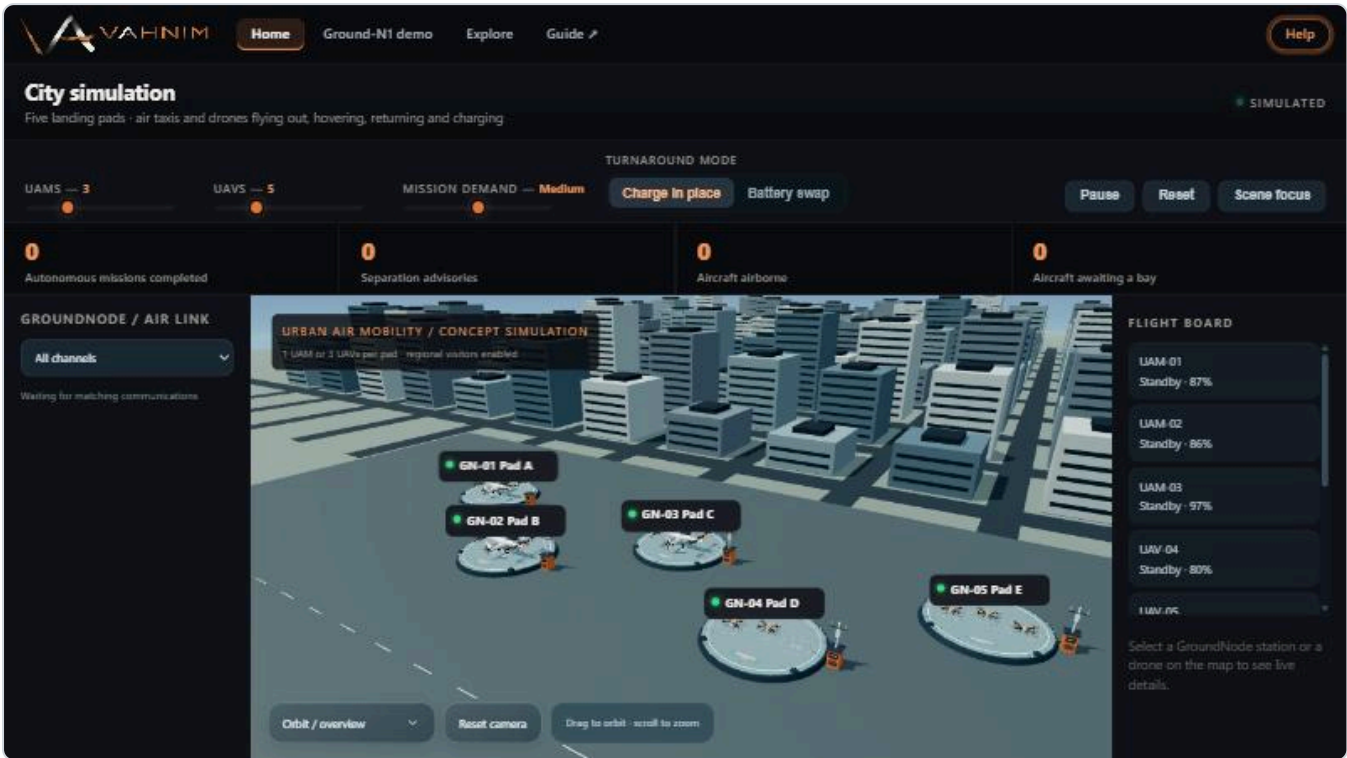


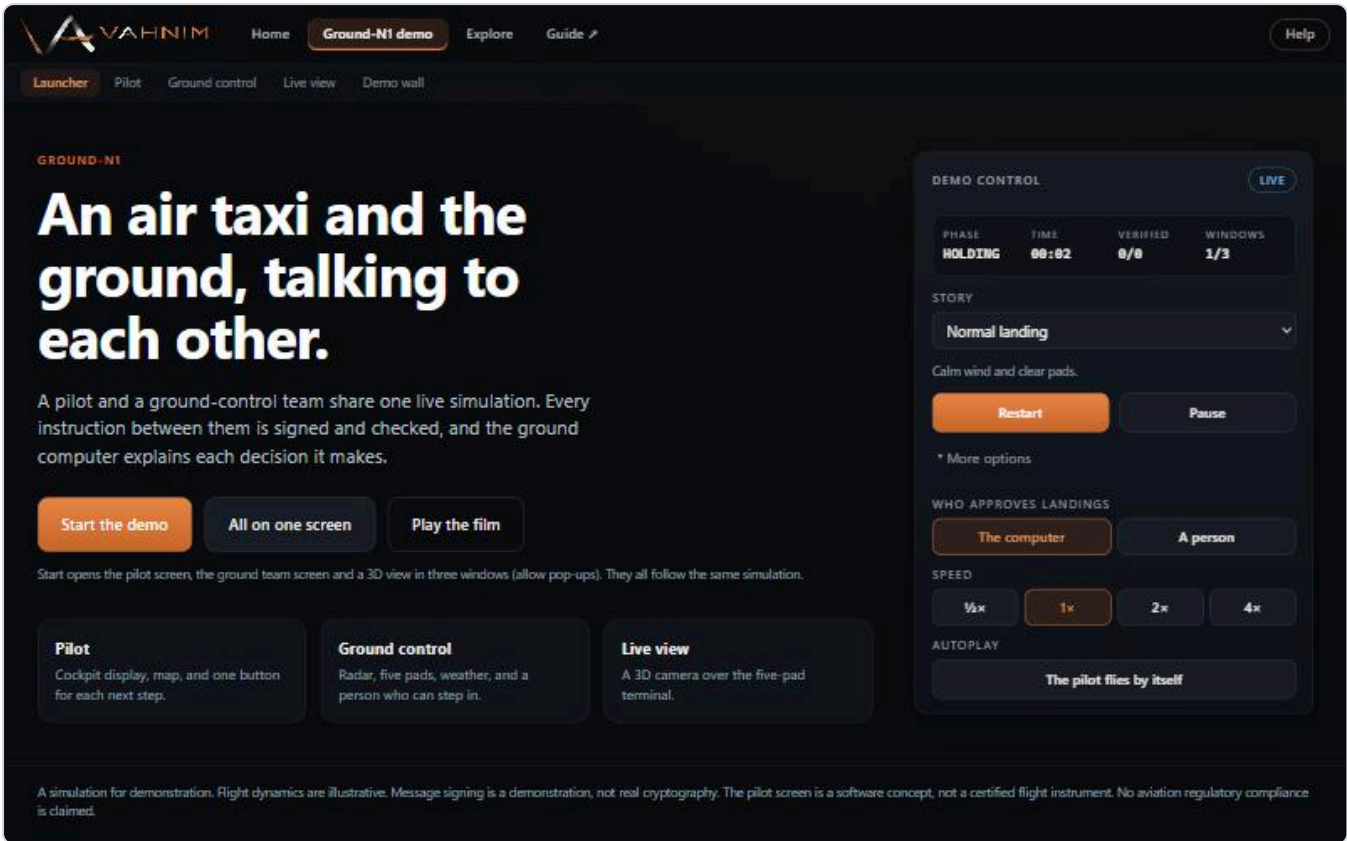
Figure 1. The five pads with their parked aircraft; controls above, flight board on the right.

Aircraft depart a pad vertically, climb, fly to a city site, **hover still** for the mission, then return, wait for a free bay, land, charge (or swap the battery) and redeploy. Each UAM takes a whole pad. Each pad holds up to three UAVs, never mixed with a UAM.

Control	What it does
UAMs (0 to 12) and UAVs (0 to 20)	How many air taxis and drones are in the home fleet. Added aircraft arrive from outside the city; removed ones retire after landing.
Mission demand	How often idle aircraft are sent out. Higher demand means busier pads and queues.
Turnaround mode	<i>Charge in place</i> (longer stop) or <i>Battery swap</i> (short stop).
Pause, Reset, Scene focus	Freeze the simulation (the camera still works), restart it, or hide the panels for a clean picture.
Camera menu	<i>Orbit</i> , <i>City overview</i> , <i>Follow aircraft</i> , <i>Pad approach</i> . Drag to orbit and scroll to zoom; <i>Reset camera</i> returns to the start view.
Flight board	Click an aircraft to select it and see its mission, battery, altitude and traffic status.
Communications filter	Show all channels, or only missions, safety, arrivals, departures, service, or the selected aircraft or pad.

Watch "Aircraft awaiting a bay" rise when you raise demand while few pads are free.

5. Ground-N1 demo



Starting

- 1. Open **Ground-N1 demo > Launcher**.
- 2. Pick a **Story** (see section 7). **More options** lets you choose who approves landings, the speed (1/2× to 4×) and autoplay.
- 3. Choose how to view it:
 - **Start the demo** opens three windows: live view, pilot and ground control (allow pop-ups).
 - **All on one screen** tiles the three screens on one display (the demo wall).
 - **Play the film** runs a complete landing and take-off by itself with no menus, ideal for screen recording. Press `Esc` to leave.

The control panel shows the aircraft's phase, the simulated time, how many messages were verified and how many windows are connected. **Restart** resets the story; **Pause** freezes flight and timers.

The three screens

Screen	Who	What it shows
Pilot	The air taxi pilot	Flight display, terminal map, one guidance card with a single main button, and the verified message log.
Ground control	The ground team	Radar, aircraft strip, five pad controls, weather, a decision card when a person must approve, and two logs.
Live view	Everyone	A 3D camera over the terminal with captions for key events.

Pilot screen

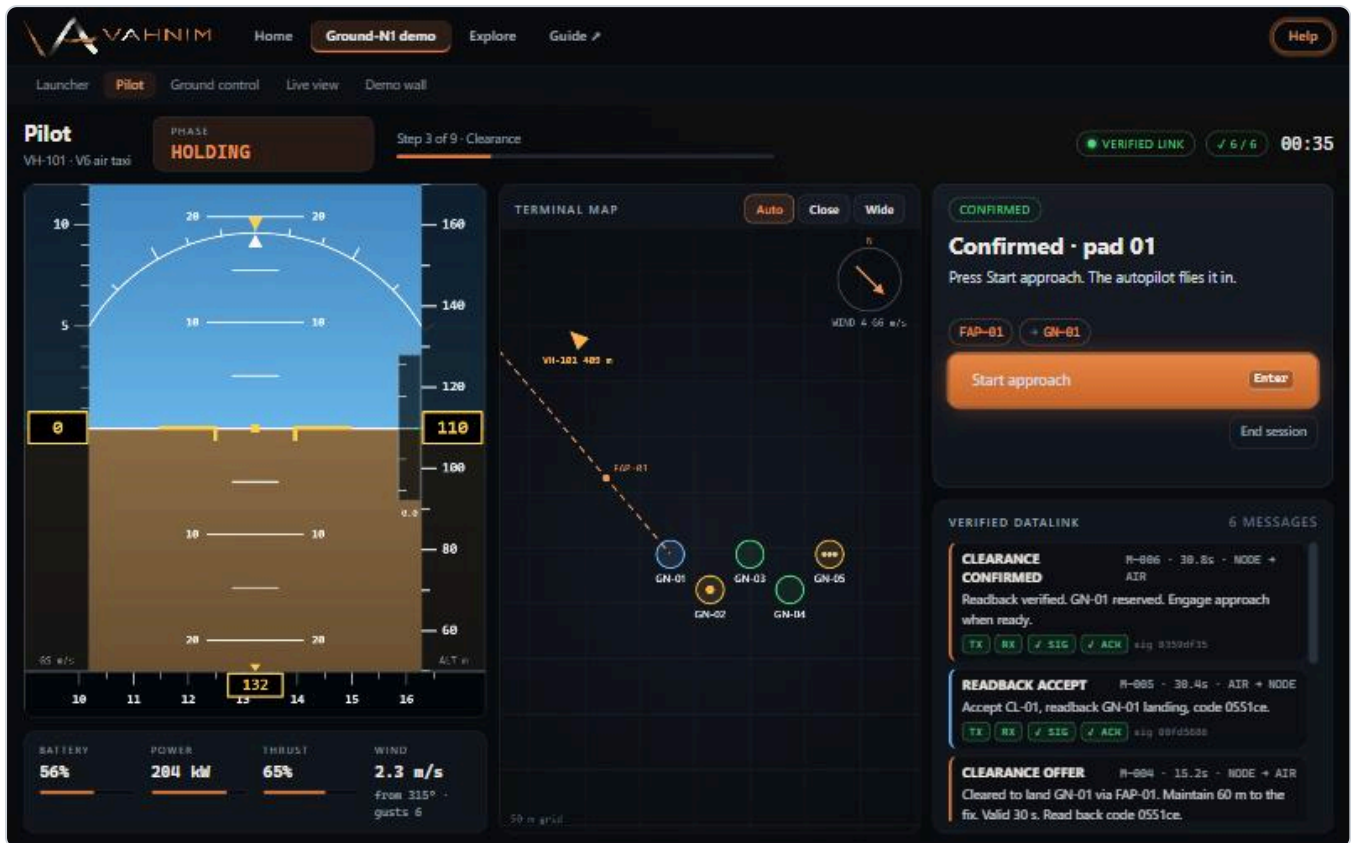


Figure 3. Pilot screen after the ground confirmed the read-back. The orange button is the next step.

- **Guidance card (right):** says what is happening and what to do next. The large button is the only thing you usually need. **Enter** presses it.
- **Decline** appears when an offer can be declined; **Go around (G)** while approaching; **End session** closes the link.
- **Flight display (left):** attitude, speed, altitude, vertical speed and heading, with battery, power, thrust and wind below.
- **Terminal map (centre):** the five pads, the route and the aircraft. Zoom with Auto, Close or Wide.
- **Verified messages:** each shows TX (sent), RX (received), SIG (signature checked) and ACK (acknowledged). A rejected message is red with the reason.

Ground control screen

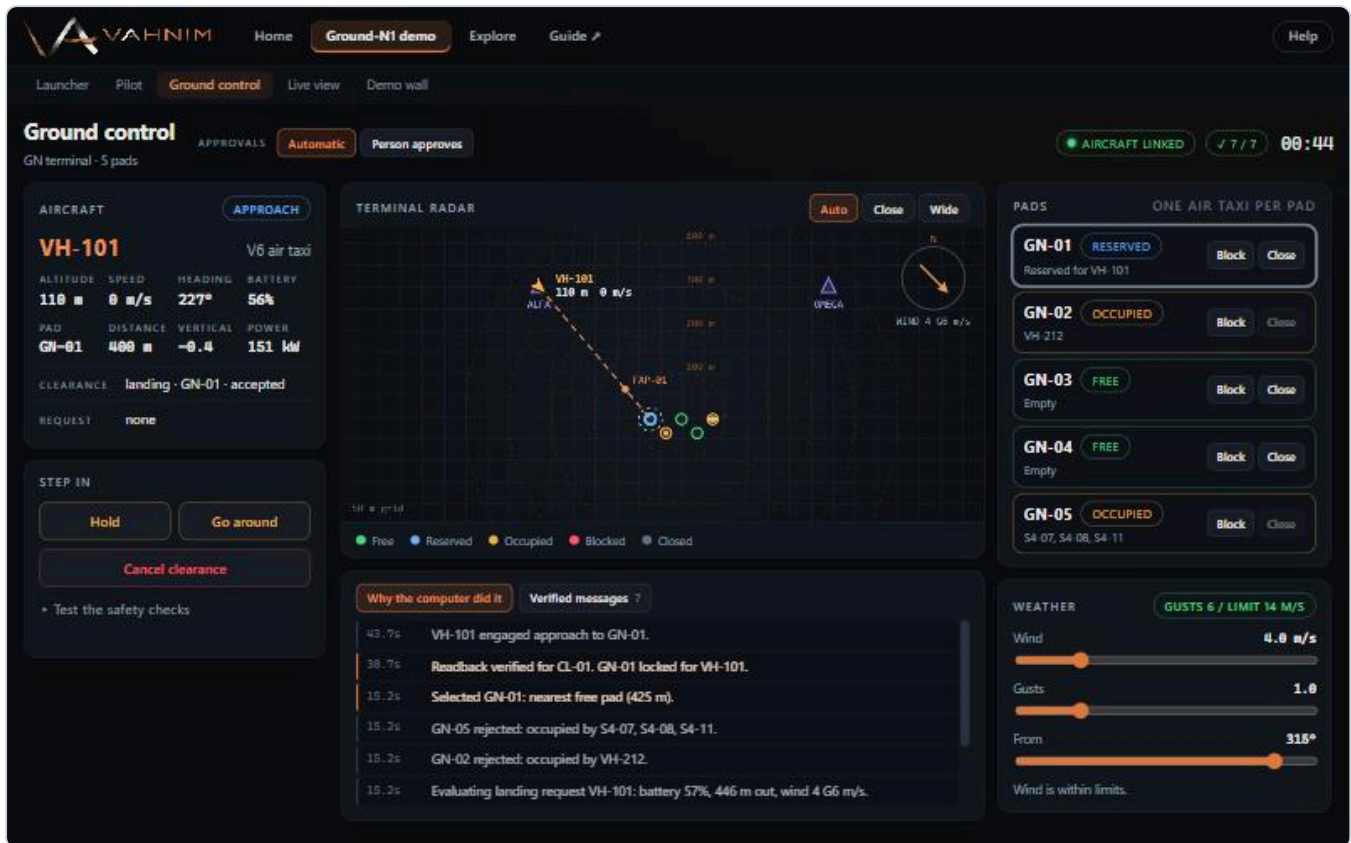


Figure 4. Ground control with the aircraft on approach to GN-01.

Element	Use
Approvals (top)	Automatic: the computer offers clearances by itself. Person approves: each offer waits for you.
Your decision card	Appears when approval is needed. Approve the recommended pad, choose another free pad, or hold.
Step in	<i>Hold</i> , <i>Go around</i> or <i>Cancel clearance</i> . These override the computer and are logged.
Pads	Each pad row has <i>Block</i> (debris) and <i>Close</i> (maintenance). A blocked or closed pad cancels an approach to it and the aircraft is sent to another pad.
Weather	Wind, gusts and direction. Above 14 m/s gusts, new requests are held.
Why the computer did it	Plain reasons for every choice: why pads were rejected, why one was picked, holds and diverts. The other tab lists the verified messages.
Test the safety checks	Send a fake message, replay a message, or drop the radio link for 10 s. Fake and replayed messages must be rejected without changing anything.

Live view

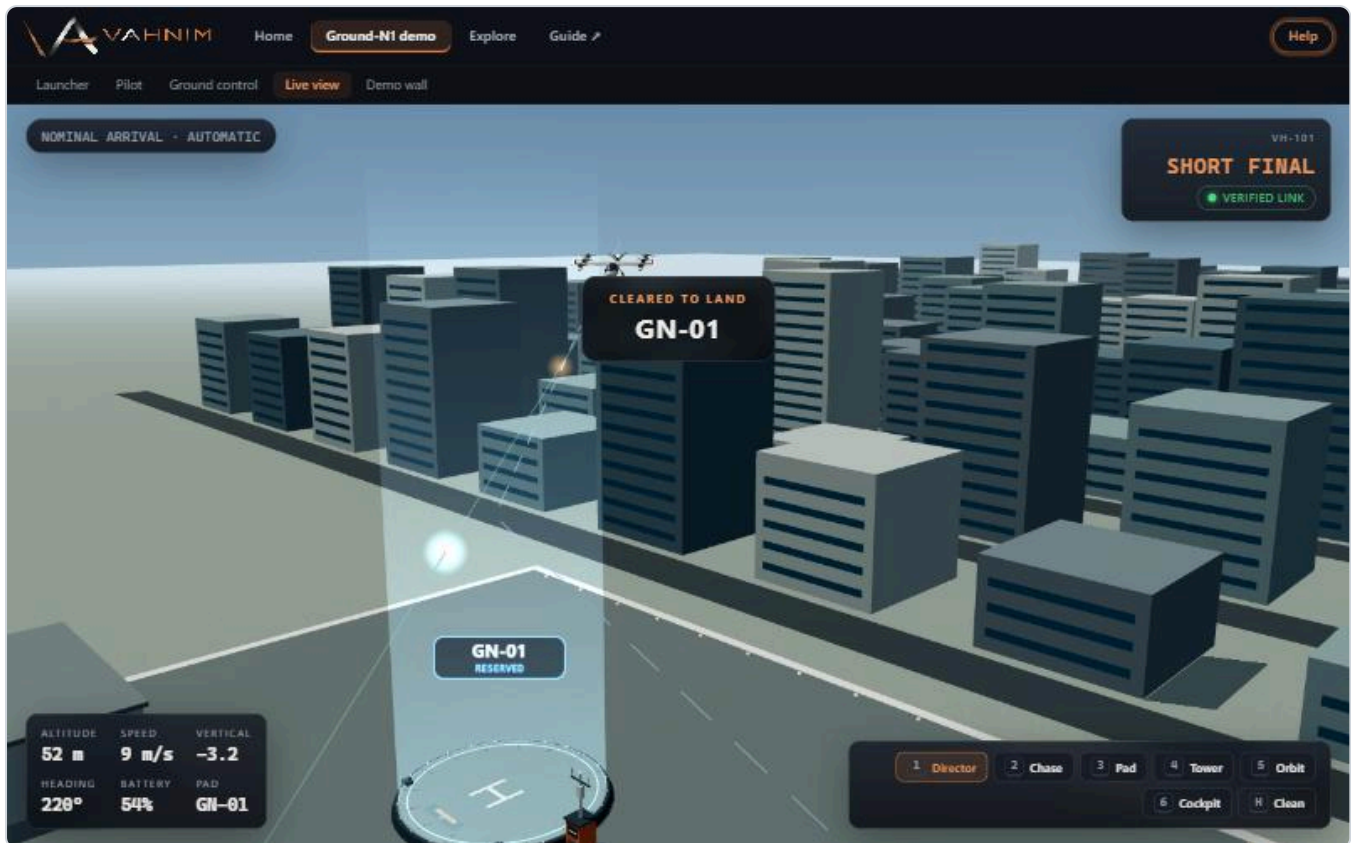


Figure 5. Live view on short final: the beam marks the assigned pad and a caption announces the clearance.

Pad colours: **green** free, **blue** reserved, **amber** occupied, **red** blocked, **grey** closed. Cameras: **1** Director (automatic shots), **2** Chase, **3** Pad, **4** Tower, **5** Orbit (drag and scroll), **6** Cockpit (first person). **H** hides the panels and header for a clean feed; **F** toggles fullscreen.

6. Walkthrough: one landing, step by step

Use the **Normal landing** story at 1×. A full run takes about two and a half minutes. Progress shows under the pilot's title.

#	Who	Action	What happens
1	Pilot	Connect to ground	A signed hello is sent; the node checks the aircraft against its registry and opens a session.
2	Pilot	Request landing	The computer checks all five pads, the wind and the battery. Occupied pads are rejected with the reason; it picks the nearest free pad.
3	Pilot	Accept and read back	The read-back code is verified; the pad is locked and shows as reserved.
4	Pilot	Start approach	The aircraft flies the route, lines up over the pad and descends. Watch the live view.
5	Automatic	Touchdown	The touchdown report and sink rate are verified. Rotors slow down.
6	Pilot	Confirm rotors stopped	Charging starts (sped up for the demo).
7	Pilot	Request departure, accept, start departure	Offered once battery and weather allow.
8	Automatic	Climb-out	The pad is released when the aircraft is clear; the session closes.

Message signing in the demo is a keyed check to show verification. It is not real cryptography.

7. Stories, approvals and safety tests

Story	What to show
Normal landing	The full happy path.
Strong wind	The request is held. On the ground screen lower the wind slider; the computer decides again and offers a clearance by itself.
Something on the pad	Debris appears on the assigned pad just before landing: the clearance is cancelled, the aircraft goes around and is offered another pad.
Low battery	The request is marked urgent and skips approval.
Radio link drops	The link drops on approach: the aircraft goes around and holds. When the link returns, connect again and re-request.

Things to try during any run

- Set approvals to **Person approves** and pick a different pad than the one recommended.
- On the ground screen, **Block** the assigned pad mid-approach.
- Open *Test the safety checks* and send a fake message: it is rejected and nothing changes.
- Set the wind above the limit on final: the computer orders a go-around.

Presenting

- For a hands-free loop use **Play the film**, or **All on one screen** with Autoplay; use 2× or 4× to shorten waits.
- Put the live view on the big screen and press  for a clean picture.

8. Explore

Overview

A plain-language menu of the pages below, each with what it is, what to try and who it is for, plus a short glossary (UAM, clearance, read-back, go-around and more).

Guided encounter

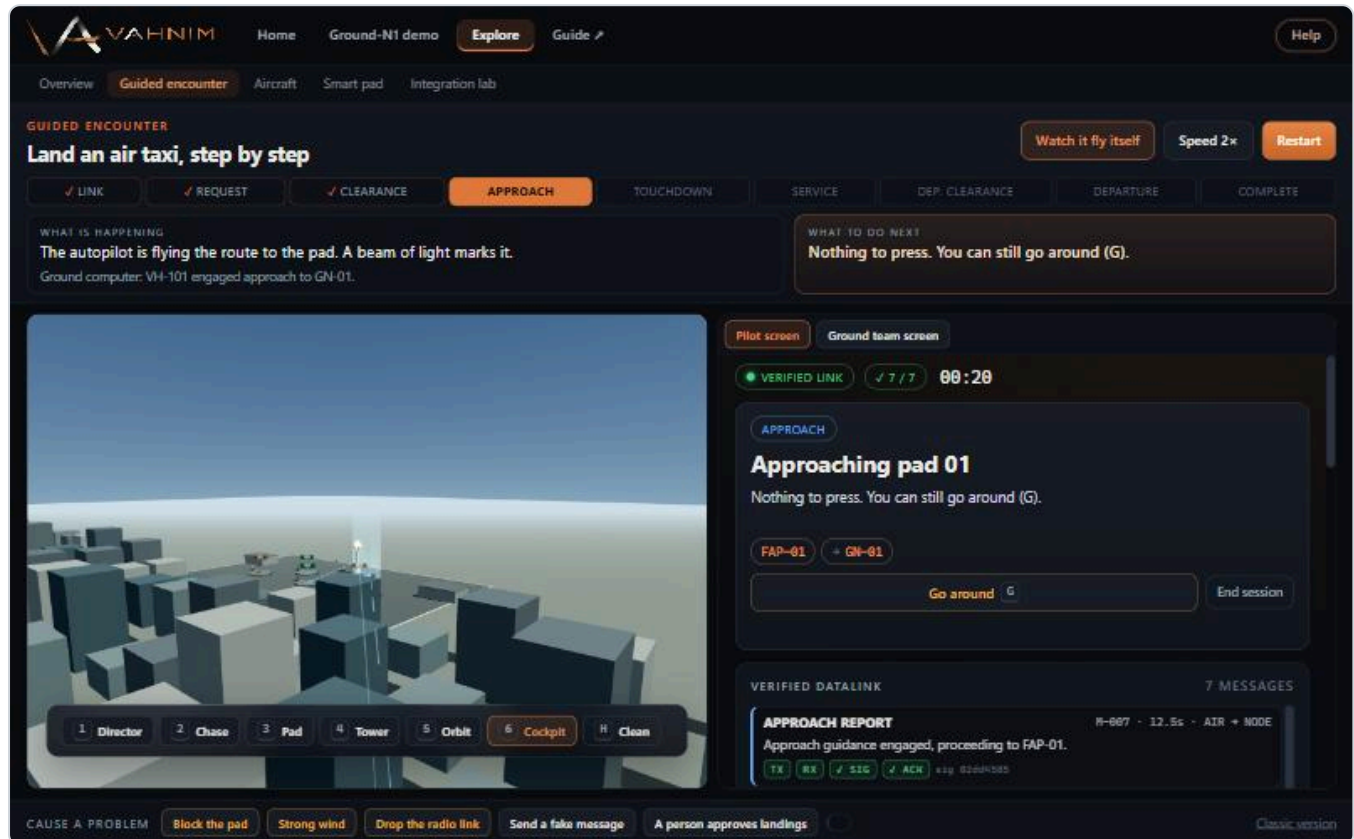


Figure 6. The guided encounter: coach on top, cockpit view and pilot screen below.

The easiest way to understand the whole system. It runs its own private simulation, so it never disturbs a demo running in other windows.

- **Coach (top):** progress steps, *What is happening* in simple words, *What to do next*, and the last decision the ground computer made.
- **Left:** a first-person view from the cockpit seat. Use the camera buttons (or keys 1 to 6) for other views.
- **Right:** the Pilot screen or, using the tabs, the Ground team screen: the same software as the demo.
- **Cause a problem (bottom):** block the pad, strong wind, drop the radio link, send a fake message, or make a person approve landings. Press again to undo.
- **Top right:** *Watch it fly itself*, *Speed* and *Restart*. The earlier version is kept under *Classic version*.

Aircraft

Turntable view of the V6 air taxi and S4 drone with plain-language key facts. Choose **Hover**, **Cruise** (the air taxi's wing carries it and the lift rotors stop; the drone tilts forward) or **Parked**.

Smart pad

Interactive 3D pad concept. Click a part or the list to highlight it; use Top view, Reset, auto-rotate and the exploded view.

Integration lab

A workshop for harder cases between aircraft and ground: visitor and unknown aircraft, expired offers and lost links. It can also mirror city telemetry from another tab.

9. Troubleshooting and limits

Problem	Fix
Page will not load	Check <code>npm start</code> is running and use <code>http://localhost:8123</code> .
Windows do not open	Allow pop-ups for the site, open the screens from the Ground-N1 menu, or use All on one screen .
A banner says "Connecting" or "Connection lost"	The screen lost the shared simulation. It reconnects by itself; if not, reload the page. If all Ground-N1 windows were closed the simulation restarts.
Screens show different states	Windows must be in the same browser profile. Adding <code>?local=1</code> to a page gives it a private copy.
3D view is blank	The browser needs WebGL. Use Chrome or Edge with hardware acceleration on.
The main button is missing	Some moments need no action; the card says "Nothing to press".

Limits. This is a simulation for demonstration. Flight dynamics are illustrative, not validated aerodynamics. Only one aircraft is simulated in Ground-N1; other pads show parked aircraft. Charging is sped up. The pilot screen is a software concept, not a certified flight instrument. Registry, read-back and message signing are demonstrations. No aviation regulatory compliance is claimed.

Developer notes (pad positions, limits, stories, aircraft specifications, colours) are in the project README under "Tuning cheat-sheet".